

ERDENI KHATUNOV

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EXPERIENCE

Yandex

- *Senior Software Engineer*

Moscow, Russia

Sep. 2023 - Sep. 2025

C++ Golang Userver PostgreSQL YTsaurus Distributed Systems High-load Backend Concurrency
Data Infrastructure

Designed and developed core distributed services for large-scale order tracking and monitoring systems.

Architected and implemented high-load microservices in C++ following Domain-Driven Design (DDD) principles, ensuring clear domain boundaries and scalability.

Designed system components including message-driven pipelines, asynchronous processing flows, and distributed MPMC queues to handle high-throughput workloads.

Owned service infrastructure end-to-end: database design (PostgreSQL, MongoDB), service deployment, performance tuning, and reliability improvements.

Built and optimized monitoring and alerting systems, including direct integration with Telegram via MTPROTO for low-latency incident notifications.

Improved system performance and fault tolerance through efficient concurrency models, resource management, and load distribution strategies.

Yandex

- *Machine Learning Engineer Intern*

Moscow, Russia

Jun. 2022 - Sep. 2022

BERT DSSM LLM Labeling Pipelines SQL

Developed and optimized AI-driven solutions for International Search.

Built offline and online e-commerce query classifiers, improving search relevance and user experience.

Created a Google parser and enrichment tool to capture the full SERP and highlight object answers, achieving competitive performance in Side-by-Side (SbS) tests.

Designed and implemented a labeling pipeline for evaluating key offline metrics for object answer criteria.

Optimized prompts and refined object answer instructions, substantially improving quality and enabling effective BERT model training.

EDUCATION

East China Normal University (ECNU)

- *Chinese Language and Culture (Non-degree Program)*

Shanghai, China

Mar. 2026 - Jul. 2026

- Intensive Chinese language training (Mandarin), including speaking, reading, and academic writing
- Coursework focused on Chinese culture, communication, and academic language proficiency
- Developed cross-cultural communication skills in an international academic environment

ITMO University AI Talent Hub

- *M.Sc. in Artificial Intelligence*

Saint Petersburg, Russia

Sep. 2025 - Present

- Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Computer Vision, MLOps, Probabilistic Models
- Focus on scalable AI systems, distributed machine learning, and foundation model development
- Applied modern ML techniques to real-world problems involving large-scale data processing, model training, and deployment

Financial University under the Government of the Russian Federation

- *B.Sc. in Applied Mathematics and Computer Science*

Moscow, Russia

Sep. 2021 - Jun. 2025

- Coursework: Algorithms and Data Structures, Discrete Mathematics, Probability Theory, Mathematical Statistics, Linear Algebra, Optimization Methods
- Specialized in distributed systems, backend engineering, and high-performance computing
- Led a team project on distributed financial systems, focusing on scalability, fault tolerance, and data consistency

Yandex School of Data Analysis (YSDA)

- *Big Data Infrastructure*

Moscow, Russia

Sep. 2023 - Dec. 2024

- Highly selective program (top-tier CS/ML school in Russia, affiliated with Yandex)
- Coursework: Distributed Systems, Database Systems (Part I & II), Concurrency, Operating Systems, C++ (Part I & II), Golang
- Focus on high-load systems, parallel computing, and production-grade ML infrastructure

PROJECTS

- P2P Binance/ByBit Telegram Bot (link)**

Python Golang TelegramAPI MTProto Pyrogram GoTD Sanic

Telegram bot was implemented to interact with the main Telegram API, capable of retrieving data from crypto exchanges. The bot can send live cryptocurrency rates and calculate potential spreads.

Jun. 2023 – Sep. 2023
- Legal text classification model (link)**

Python Transformers SkLearn TensorFlow Torch

Fine-tuned BERT language model for legal text classification.

Apr. 2023 – Jun. 2023
- Real Estate valuation forecast model (link)**

Python Pandas NumPy LightGBM Multi-Armed bandits algorithm

Developed a solution for prognosticating real estate valuations, leveraging the intricate intricacies of a multi-armed bandit algorithm. Implemented a sophisticated gradient boosting model to emulate demand dynamics at each iteration of the algorithm.

Oct. 2022 - Nov. 2022

TECHNICAL SKILLS

- Programming Languages:** C++17/20/23, Go, Python (NumPy, Pandas, asyncio), SQL (PostgreSQL, MySQL), Bash
- Backend Development:** Userver, Gin Gonic, REST API, gRPC, Protobuf, WebSockets, High-load microservices, Distributed systems
- Data & Machine Learning:** TensorFlow, PyTorch, HuggingFace Transformers, OpenCV, scikit-learn, NLP, Computer Vision, Feature engineering, Model deployment (ONNX, TensorRT)
- DevOps & Cloud Platforms:** Docker, Kubernetes (OpenShift, Istio Service Mesh), AWS S3, Kafka, GitLab CI/CD, Helm, Terraform, Linux server administration
- Databases & Storage:** PostgreSQL, MongoDB, ClickHouse, YDB, YT, Redis, TimescaleDB
- Additional Skills:** API design, Distributed transaction processing, Time synchronization in distributed systems, System performance optimization, Code review & team collaboration

ACHIEVEMENTS

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|---|------------------------------|
| 2022 The ICPC Northern Eurasia Finals | Contestant |
| 2020, 2021 The All-Russian School Team Programming Olympiad | Diploma of the second degree |
| 2020 Moscow Olympiad in informatics | Prize Winner |

PUBLICATIONS AND RESEARCH

- AI Technologies in Service of Family and Society**
Speech at XV International Scientific Student Congress, hosted by Financial University
- Digitalization of National Currency as a Global Trend**
Report at International Scientific Student Conference

OTHER ACTIVITIES

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| MIPT Olympiad Schools | Moscow, Russia |
| <i>Algorithms and data structures.</i> | Jul. 2019 |
| Sirius | Sochi, Russia |
| <i>Algorithms and data structures.</i> | Oct. 2019 |

LANGUAGES

- English:** IELTS Academic 7.5 (C1)
- Chinese (Mandarin):** Elementary proficiency (currently studying at ECNU)
- Russian:** Native
- Buryat:** Native